

## Comparison of NoSQL databases

| Database              | Developer                              | Licence                                 | Type                                                                                                                        | Language                                      | Description                                                                                                                                                                             |
|-----------------------|----------------------------------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudant              | Originally by Cloudant Inc, now by IBM | Proprietary                             | NoSQL, DBaaS                                                                                                                | Erlang, Java, C, Scala                        | Open source non-relational distributed database. Based on CouchDB, open source BigCouch, an Apache-backed project.                                                                      |
| Cassandra             | Apache Software Foundation             | Apache licence                          | NoSQL database                                                                                                              | Java                                          | Open source distributed database management system.                                                                                                                                     |
| BaseX                 | BaseX Team                             | BSD licence                             | XML database                                                                                                                | Java, XQuery                                  | XML database management system. Provides support for XQuery and XPath.                                                                                                                  |
| Clusterpoint database | Clusterpoint Ltd                       | Free licence                            | Distributed database, Enterprise search operational database, Document-oriented NoSQL, XML, JSON, SQL database, Cloud DBaaS | C, C++, SQL, PHP, Java, .NET, Python, Node.js | Document-oriented database server platform used for the storage and processing of XML and JSON data. Enables users to perform the transactions in the same way as with SQL databases.   |
| MongoDB               | MongoDB Inc.                           | GNU AGPL v3.0 (drivers: Apache licence) | Document-oriented database                                                                                                  | C, C++, JavaScript                            | Cross-platform document-oriented database. Classified as a NoSQL database.                                                                                                              |
| Lotus Notes           | IBM                                    | Proprietary                             | Collaborative software, personal information manager, email client                                                          | Java/ Eclipse, C++                            | Business collaboration functionality such as file sharing, micro-blogging, instant messaging, to-do lists, contacts, emails, calendars, discussion forums, blogs, and user directories. |
| Orient DB             | Orient Technologies                    | Apache licence                          | Document-oriented database, Graph database, Multi-model database                                                            | Java                                          | Open source NoSQL database management system. Relationships are managed as in graphs with direct connections.                                                                           |

and low latency to read and write is required.

3. It is preferred when there is a tight connection between the data.
4. Cassandra is used when there is a need to store a large amount of data feeds from customers.
5. It is used to store a large volume of data by compressing it, e.g., sound files of about 120MB can be stored by compressing them to about 2-5MB.
6. It can be used for persistent data storage, i.e., data will be available only for the specified session, and for finding particular data based on the situation.
7. It can be used for geographically distributing the data and for real-time querying.

## Case study

There are various organisations that are using Apache Cassandra to maintain data distribution across various servers. Let us look at how these organisations switched to this technology in the following case study of eBay.

eBay is one of the world's largest online markets that enables its users to buy and sell anything, from anywhere. Whenever any user requests for a search, a huge amount of data corresponding to this search needs to be retrieved and

displayed on the screen. There was a need for a database that enabled retrieval with high speed, to satisfy the customer's requirements.

Cassandra can be used in such cases as it supports a large amount of data across various nodes. It makes data retrieval from these nodes easy, provides low latency, high throughput, and ensures all-time availability even during peak loads as well as accuracy of the searched data, which satisfies the customer. 

## References

- [1] [http://en.wikipedia.org/wiki/Document-oriented\\_database](http://en.wikipedia.org/wiki/Document-oriented_database)
- [2] <http://www.cubrid.org/blog/dev-platform/nosql-benchmarking/>
- [3] <http://en.wikipedia.org/wiki/NoSQL>
- [4] [http://en.wikipedia.org/wiki/Apache\\_Cassandra](http://en.wikipedia.org/wiki/Apache_Cassandra)
- [5] <http://www.datastax.com/resources/casestudies/eBay>
- [6] <http://planetcassandra.org/apache-cassandra-use-cases/>

## By: Priyanka Agashe and Mitesh Soni

Priyanka Agashe is a senior software engineer. Her professional interests include testing, Oracle apps, electronics and robotics.

Mitesh Soni is a technical lead. He loves to write about new technologies. Blog: <http://clean-clouds.com>